Prof. Dr. W. A. ZÚÑIGA-GALINDO

Debnath Endowed Professor in Mathematical Sciences
The University of Texas Rio Grande Valley
School of Mathematical and Statistical Sciences, United States.
1201 West University Drive, Edinburg, TX 78539, United States
[email] • [phone]

Web Pages

Arxiv; Research Gate; Google Scholar; MATHSCINET; Mathematics Genealogy Project; ScholarGPS, ORCID.

Research Profile

I work at the interface of non-Archimedean (p-adic) analysis, partial differential equations, mathematical physics, and, more recently, the mathematical foundations of deep learning. My research program has developed a systematic theory of pseudodifferential operators and parabolic-type equations over p-adic and adelic spaces, and has applied this framework to Markov processes, models of complex hierarchical systems in biology, string amplitudes and Koba–Nielsen integrals in quantum field theory, and, over the last several years, a rigorous non-Archimedean/statistical-field-theoretic formulation of deep neural networks. I have authored 3 research monographs (Cambridge University Press, Springer, De Gruyter) and more than 80 papers in journals spanning mathematics and mathematical physics (*Advances in Mathematics*, *J. Fourier Anal. Appl.*, *Trans. Amer. Math. Soc.*, *Reviews in Mathematical Physics*, *Nuclear Physics B*, *J. Phys. A*, *Annals of Physics*, *Phys. A: Stat. Mech. Appl*, *Phys. D: Nonlinear Phenom.*, *J. High Energy Phys.*, *Nonlinearity*), supervised 17 Ph.D. dissertations in Colombia, Mexico, and the United States, and served on the editorial boards of six international journals.

Citation metrics: ∼2,000 citations on Google Scholar (August 2026); full citation history and h-index available at my Google Scholar profile.

Education

- D.Sc. Mathematics, 1996, National Institute of Pure and Applied Mathematics, IMPA, Rio de Janeiro, Brazil.
- M.Sc. Mathematics, 1992, Los Andes University, Bogotá D.C., Colombia.
- M.Sc. Electrical Engineering, 1992, Los Andes University, Bogotá D.C., Colombia.
- B.Sc. Mathematics, 1988, National University of Colombia, Bogotá D.C., Colombia.
- B.Sc. Electronics Engineering, 1988, Francisco José de Caldas University, Bogotá D.C., Colombia.

Appointments

- (12/2020 to present) Full Professor of Mathematics with tenure, the University of Texas Rio Grande Valley, United States.
- (2023-2025) Doctoral Faculty, Department of Physics, the University of Texas Rio Grande Valley, United States.
- (04/2016-12/2020) Research Professor level 3D at the Department of Mathematics, Center for Research and Advanced Studies of the National Polytechnic Institute, CINVESTAV, Mexico.
- (01/2013- 03/2016) Research Professor level 3C at the Department of Mathematics, Center for Research and Advanced Studies of the National Polytechnic Institute, CINVESTAV, Mexico.
- (05/2007-12/2013) Research Professor level 3B at the Department of Mathematics, Center for Research and Advanced Studies of the National Polytechnic Institute, CINVESTAV, Mexico.
- (03/2006- 04/2007) Associate Professor of Computer Science and Mathematics, Barry University, Department of Mathematics and Computer Science, Miami, U.S.A.

- (09/2001- 02/2006) Assistant Professor of Computer Science and Mathematics, Barry University, Department of Mathematics and Computer Science, Miami, Florida, U.S.A.
- (10/1997-09/2000) Director of Scientific Computation Laboratory, Autonomous University of Bucaramanga, Colombia.
- (08/1996-09/1997) Assistant Professor of Mathematics, Industrial University of Santander, Department of Mathematics, Bucaramanga, Colombia.
- (09/1991-09/1992). Research Engineer at Institute of Electronics and Communications, ITEC-TELECOM, Bogotá, Colombia.

Awards

- (1/09/2026-31/08/2029) Reappointment - Debnath Endowed Professorship in Mathematical Sciences, the University of Texas Rio Grande Valley.
- (1/09/2023-31/08/2026) Reappointment - Debnath Endowed Professorship in Mathematical Sciences, the University of Texas Rio Grande Valley.
- (12/14/2020- 31/08/2023) Debnath Endowed Professorship in the Mathematical Sciences, the University of Texas Rio Grande Valley.
- (2021-2025) Member of National System of Researchers of Mexico, SNI, Investigador Nacional Nivel III (highest possible level).
- (2016-2020) Member of National System of Researchers of Mexico, SNI[1], Investigador Nacional Nivel III (highest possible level).
- (2014) Regular Member of the Mexican Academy of Sciences.
- (2012-2015) Member of the National System of Researchers of Mexico, SNI, Investigador Nacional Nivel II.

[1]The National Systems of Researchers of Mexico is a Mexican version of the Canada Research Chairs.

- (2010) Alejandro Ángel Escobar Medal in Exact Physical and Natural Sciences, Colombia.
- (2009) Regular Member of the Colombian Academy of Exact, Physical and Natural Sciences, Bogotá D.C., Colombia.
- (2007-2011) Member of the National System of Researchers of Mexico, SNI, Investigador Nacional Nivel I.
- (2004-2005) Sister Jeanne O'Laughlin Scholar's Award, Barry University, Miami, Florida, U.S.A.

Editorships

- (2025 to present) Member of the Editorial Board of Scientific Reports.
- (2024 to present) Member of the Editorial Board of Axioms.
- (2021 to present) Member of the Editorial Board of the Turkish Journal of Mathematics.
- (2021 to present) Member of the Editorial Board of the Journal of Nonlinear Mathematical Physics.
- (2016 to present) Member of the Editorial Board of p-Adic Numbers, Ultrametric Analysis, and Applications.
- (2011 to present) Editor of Revista Colombiana de Matematicas. This journal is published by the Colombian Mathematical Society.

Grants

- Multiple grants for organizing meetings, conferences, and workshops in France, Germany, Spain, Mexico, and Colombia are not listed here. Please see the list of conferences that I have co-organized.
- (2020-2024) Group grant from Conacyt, Mexico, to conduct research in mathematical biology (Non-conventional mathematical and computational models for the study and analysis of relevant problems in biology). Multidisciplinary research grant.

- (2016-2020) Grant from Conacyt, Mexico, to conduct research in number theory/algebraic geometry, p-adic analysis and mathematical physics (Non-Archimedean Analysis and Geometry and Applications).
- (2011-2014) Grant from Conacyt, Mexico, to conduct research in number theory, algebraic geometry and p-adic analysis (Oscillatory Integrals and Exponential sums in Arithmetic and Complex Systems).
- (2005-2007) Young Investigator Grant, National Security Agency, US, to conduct research in number theory and p-adic analysis (Newton polyhedra, Oscillatory Integrals and Pseudo-Differential Operators Over p-adic Fields).
- Travel Grant from American Institute of Mathematics to attend the Workshop "Numerical Invariants of Singularities and Higher-dimensional Algebraic Geometry," Palo Alto, California, US, July 31st to August 4th, 2006.
- (2003-2004) Grant from Council of Grants and Research, Barry University, Miami, U.S.A, to conduct research in number theory and p-adic analysis (Local zeta functions and p-adic pseudo-differential equations).
- (2000-2002) Grant from COLCIENCIAS (Administrative Department of Science, Technology and Innovation), Colombia, to conduct research in number theory (Local Zeta Functions, non Degeneracy Conditions and Newton Polygons).
- (1998-2000) Grant from COLCIENCIAS (Administrative Department of Science, Technology and Innovation), Colombia, to conduct research in number theory (Local Zeta Functions of quasi-homogeneous polynomials).

Visiting Positions

- Visiting Professor, Department of Mathematics, Howard University, Washington D.C., US, April 5-12, 2024.
- Visiting Professor, Department of Mathematics, Pontifical Xavierian University, and Department of Mathematics, National University of Colombia, Bogotá D.C., Colombia, March 11-15, 2024.

- Visiting Professor, Department of Mathematics, University of Leuven, KU Leuven, Belgium, June 3rd - 17th, 2017.
- Visiting Professor, Department of Mathematics, University of Concepción, Chile, January 16th - 25th, 2017.
- Visiting Professor, Department of Mathematics, Pontifical Xavierian University, Bogotá D.C., Colombia, April 3rd-7th, 2017.
- Visiting Professor, Department of Mathematics, University of del Valle, Cali, Colombia, May 17th-20th, 2015.
- Visiting Professor, Department of Mathematics, University of Leuven, KU Leuven, Belgium, April 9th-12th, 2013.
- Visiting Professor, Department of Mathematics, National University of Colombia, Bogotá, November 26th to December 18th, 2012.
- Visiting Professor, Department of Mathematics, National University of Colombia, August 3rd -14th, 2009.
- Visiting Professor, Department of Mathematics, University of Leuven, KU Leuven, Belgium, July 1st - 20th, 2009.
- Visiting Professor, Department of Mathematics, National University of Colombia, Bogotá, June 21st to 29th, 2008.
- Visiting Professor, Institute of Mathematics and Computation, São Paulo University, ICMC-USP, campus São Carlos, Brazil, July 4th to August 2nd, 2008.
- Visiting Professor, Department of Mathematics, University of Leuven, KU Leuven, Belgium, June 2004.
- General Membership, Summer Program 2002, Mathematical Sciences Research Institute, MSRI, Berkeley, July 2002.
- Visiting Professor, Institute of Mathematics and Computation, São Paulo University, ICMC-USP, campus São Carlos, Brazil, January 2000.
- Visiting Professor, Department of Mathematics, University of Leuven, KU Leuven, Belgium, January 1999.

- Visiting Professor, Graduate Center and Baruch College, The City University of New York, January 1998.
- Visiting Professor, Institute of Pure and Applied Mathematics, IMPA, January 1997.

Books Edited

1. p-Adic Analysis, Arithmetic and Singularities- Carlos Galindo, Alejandro Melle-Hernández, Julio José Moyano-Fernández, Wilson A. Zúñiga-Galindo, Editors. Contemporary Mathematics, 2022.
2. Advances in Non-Archimedean Analysis and Applications. The p-adic Methodology in STEAM-H. STEAM-H: Science, Technology, Engineering, Agriculture, Mathematics & Health. Edited by W. A. Zúñiga-Galindo and T. Bourama. Springer, Cham 2021.
3. Zeta Functions in Algebra and Geometry - Antonio Campillo, Universidad de Valladolid, Gabriel Cardona, Universitat de les Illes Balears, Alejandro Melle-Hernández, Universidad Complutense de Madrid, Wim Veys, University of Leuven, and Wilson A. Zúñiga-Galindo, Cinvestav I.P.N., Editors - AMS, Contemporary Mathematics 2012.

Books

1. Zúñiga-Galindo W. A., "p-Adic Analysis: Stochastic Processes and Pseudo-Differential Equations." De Gruyter, 2025.
2. Khrennikov Andrei, Kozyrev Sergei, Zúñiga-Galindo W. A., "Ultrametric Pseudodifferential Equations and Applications". Encyclopedia of Mathematics and its Applications 168. Cambridge: Cambridge University Press (ISBN 978-1-107-18882-2), 2018.
3. Zúñiga-Galindo W. A., "Pseudodifferential Equations Over Non-Archimedean Spaces," Lectures Notes in Mathematics 2174, Springer, 2016. ISBN 978-3-319-46738-2.

Selected Publications

(A representative subset of the full, chronologically ordered list in the "Papers Published" section below.)

1. Khrennikov Andrei, Kozyrev Sergei, Zúñiga-Galindo W. A., "Ultrametric Pseudodifferential Equations and Applications". Encyclopedia of Mathematics and its Applications 168. Cambridge University Press, 2018.

2. Zúñiga-Galindo, W. A., "p-Adic Analysis: Stochastic Processes and Pseudo-Differential Equations," De Gruyter, 2025.

3. Zúñiga-Galindo, W. A. "Deep Neural Networks: A Formulation via Non-Archimedean Analysis," J. Fourier Anal. Appl. 32 (2026), no. 4, Paper No. 77.

4. Zúñiga-Galindo, W. A. "A Correspondence Between Deep Boltzmann Machines and p-Adic Statistical Field Theories," Adv. Theor. Math. Phys. 28 (2024), no. 2, 679–741.

5. Bocardo-Gaspar M., Veys W. & Zúñiga-Galindo W.A., "Meromorphic continuation of Koba-Nielsen string amplitudes," J. High Energ. Phys. 2020, 138 (2020).

6. Veys Willem, Zúñiga-Galindo W. A., "Zeta Functions and Oscillatory Integrals for Meromorphic Functions," Adv. Math. 311 (2017), 295–337.

7. Zúñiga-Galindo W. A., "Non-Archimedean Reaction-Ultradiffusion Equations and Complex Hierarchic Systems," Nonlinearity 31 (2018), no. 6, 2590–2616.

8. Zúñiga-Galindo W. A., "Reaction-diffusion equations on complex networks and Turing patterns, via p-adic analysis," J. Math. Anal. Appl. 491 (2020), no. 1, 124239.

9. Zúñiga-Galindo, W. A., "Non-Archimedean statistical field theory," Rev. Math. Phys. 34 (2022), no. 8, Paper No. 2250022.

10. Torba S. M., Zúñiga-Galindo W. A., "Parabolic Type Equations and Markov Stochastic Processes on Adeles," J. Fourier Anal. Appl. 19 (2013), no. 4, 792–835.

11. Zúñiga-Galindo, W. A., "Local Zeta Functions Supported on Analytic Sets and Newton Polyhedra," Int. Math. Res. Not. IMRN 2009, no. 15, 2855–2898.

12. Saia M.J., and Zuniga-Galindo W.A., "Local zeta functions for curves, non-degeneracy conditions and Newton polygons," Trans. Amer. Math. Soc. 357 (2005), 59–88.

13. Zúñiga-Galindo, W. A., "The p-Adic Schrödinger equation and the two-slit experiment in quantum mechanics," Ann. Physics 469 (2024), Paper No. 169747.

Papers Published

1. Zúñiga-Galindo W. A., Zambrano-Luna B. A., and Indoung, Chayapuntika. "Pattern Formation in Quantum Hierarchical Cellular Neural Networks."Communications in Nonlinear Science and Numerical Simulation 163 (2026), part 5, Paper No. 110715.

2. Zúñiga-Galindo W. A., Zambrano-Luna B. A., and Indoung, Chayapuntika. "Pseudo-Traveling Waves and Bumps in Quantum and Classical Hierarchical Cellular Neural Networks. "Journal of Pseudo-Differential Operators and Applications 17, 65 (2026).

3. Zúñiga-Galindo, W. A., Mayes, Nathaniel P." p-Adic quantum mechanics, infinite potential wells, and continuous-time quantum walks," Reviews in Mathematical Physics (2026), Paper No. 2550027 (32 pages).

4. Zúñiga-Galindo, W. A. "Deep Neural Networks: A Formulation via Non-Archimedean Analysis," J. Fourier Anal. Appl. 32 (2026), no. 4, Paper No. 77.

5. Veys Willem, Zúñiga-Galindo W. A., "Koba-Nielsen local zeta functions, convex subsets, and generalized Selberg-Mehta-Macdonald and Dotsenko-Fateev-like integrals," Commun. Number Theory Phys. 20 (2026), no. 2, 353–390.

6. Zúñiga-Galindo, W.A., "Quantum mechanics, non-locality, and the space discreteness hypothesis," Ann. Physics 489 (2026), Paper No. 170459.

7. Zúñiga-Galindo W. A., "2-Adic Quantum Mechanics, Continuous-Time Quantum Walks, and the Space Discreteness." Fortschr. Phys. 73 (2025), no. 8, Paper No. e70019.

8. Zúñiga-Galindo W. A., "Dynamic mean-field theory for continuous random networks," 2025 J. Phys. A: Math. Theor. 58 12520.

9. Zúñiga-Galindo W. A., "A Correspondence Between Deep Boltzmann Machines and p-Adic Statistical Field Theories," Adv. Theor. Math. Phys. Volume 28, Number 2, pp. 679-741 (2024).

10. Zúñiga-Galindo W.A., Zambrano-Luna B.A. & Dibba B., "Hierarchical Neural Networks, p-Adic PDEs, and Applications to Image Processing," J Nonlinear Math Phys 31, 63 (2024).

11. Zúñiga-Galindo W. A., "The p-Adic Schrödinger equation and the two-slit experiment in quantum mechanics," Ann. Physics 469 (2024), Paper No. 169747.

12. Zúñiga-Galindo W. A., "p-Adic Quantum Mechanics, the Dirac Equation, and the violation of Einstein causality," J. Phys. A: Math. Theor. 57 305301 (2024).

13. Chacón-Cortés L. F., Garcia-Bibiano C. A., Zúñiga-Galindo W. A., "Turing Patterns in a p-Adic FitzHugh-Nagumo System on the Unit Ball," p-Adic Num Ultrametr Anal Appl 15, 245–265 (2023).

14. Zúñiga-Galindo, W. A., and B. A. Zambrano-Luna. 2023. "Hierarchical Wilson–Cowan Models and Connection Matrices," Entropy 25, no. 6: 949. https://doi.org/10.3390/e25060949.

15. Zúñiga-Galindo W. A., He. C., Zambrano-Luna B. A., "p-Adic statistical field theory and convolutional deep Boltzmann machines," Progress of Theoretical and Experimental Physics, Volume 2023, Issue 6, June 2023, 063A01, https://doi.org/10.1093/ptep/ptad061.

16. Zambrano-Luna B., Zúñiga-Galindo W. A., "p-Adic Cellular Neural Networks," J. Nonlinear Math. Phys. 30 (2023), no. 1, 3470. 11.

17. Zambrano-Luna, B. A., Zúñiga-Galindo, W. A., "p-adic cellular neural networks: Applications to image processing," Phys. D 446 (2023), Paper No. 133668.

18. Zúñiga-Galindo, W. A., "p-adic statistical field theory and deep belief networks," Phys. A 612 (2023), Paper No. 128492, 23 pp.

19. Chacón-Cortés L. F., Garcia-Bibiano C. A., Zúñiga-Galindo, W. A., "Local Well-Posedness of the Cauchy Problem for a p-Adic Nagumo-Type Equation," p-Adic Numbers Ultrametric Anal. Appl. 14 (2022), no. 4, 279–296.

20. Zúñiga-Galindo W. A., "Non-Archimedean quantum mechanics via quantum groups," Nuclear Phys. B 985 (2022), Paper No. 116021, 21 pp.

21. Zúñiga-Galindo, W. A.,"Non-Archimedean statistical field theory," Rev. Math. Phys. 34 (2022), no. 8, Paper No. 2250022, 41 pp.

22. Zúñiga-Galindo W. A., "Eigen's paradox and the quasispecies model in a non-Archimedean framework," Phys. A 602 (2022), Paper No. 127648, 18 pp.

23. Zúñiga-Galindo W. A. p-adic analysis: a quick introduction. p-adic analysis, arithmetic and singularities, 177–221, Contemp. Math., 778, Amer. Math. Soc., [Providence], RI, [2022], ©2022.

24. Zúñiga-Galindo, W. A., "Ultrametric diffusion, rugged energy landscapes and transition networks," Phys. A 597 (2022), Paper No. 127221, 19 pp.

25. Fuquen-Tibatá A. R., García-Compeán H., Zúñiga-Galindo W. A., Euclidean quantum field formulation of p-adic open string amplitudes, Nuclear Phys. B 975 (2022), Paper No. 115684, 27 pp.

26. Zúñiga-Galindo W. A., Zambrano-Luna B. A., León-Cardenal E., "Graphs, local zeta functions, log-Coulomb gases, and phase transitions at finite temperature," J. Math. Phys. 63 (2022), no. 1, Paper No. 013506, 21 pp.

27. Zúñiga-Galindo W.A., "Non-Archimedean Models of Morphogenesis." In: Zúñiga-Galindo W.A., Toni B. (eds) Advances in Non-Archimedean Analysis and Applications. STEAM-H: Science, Technology, Engineering, Agriculture, Mathematics & Health. Springer, Cham.(2021).

28. Bocardo-Gaspar M., García-Compeán H., López, E.Y., Zúñiga-Galindo W.A., "Local Zeta Functions and Koba–Nielsen String Amplitudes," Symmetry 2021, 13, 967.

29. Bocardo-Gaspar M., Veys W. & Zúñiga-Galindo W.A., "Meromorphic continuation of Koba-Nielsen string amplitudes," J. High Energ. Phys. 2020, 138 (2020).

30. Zúñiga-Galindo W. A., "Reaction-diffusion equations on complex networks and Turing patterns, via p-adic analysis," J. Math. Anal. Appl. 491 (2020), no. 1, 124239, 39 pp.

31. Zúñiga-Galindo W. A., Torba, Sergii M., "Non-Archimedean Coulomb gases," J. Math. Phys. 61 (2020), no. 1, 013504, 16 pp.

32. García-Compeán H., López Edgar Y., Zúñiga-Galindo W. A., "p-adic open string amplitudes with Chan-Paton factors coupled to a constant B-field," Nuclear Phys. B 951 (2020), 114904, 33 pp.

33. Arroyo-Ortiz Edilberto, Zúñiga-Galindo W. A., "Construction of p-adic Covariant Quantum Fields in the Framework of White Noise Analysis". Rep. Math. Phys. 84 (2019), no. 1, 1–34.

34. Bocardo-Gaspar M, Garcia-Compean H., Zúñiga-Galindo W. A., "Regularization of p-adic String Amplitudes, and Multivariate Local Zeta Functions," Lett. Math. Phys. 109 (2019), no. 5, 1167–1204.

35. Mendoza-Martínez M. L., Vallejo-Rodríguez J. A., Zúñiga-Galindo W. A., "Acausal quantum theory for non-Archimedean scalar fields," Rev. Math. Phys. 31 (2019), no. 4, 1950011, 46 pp.

36. Zúñiga-Galindo W. A., "Non-Archimedean Replicator Dynamics and Eigen's Paradox,"J. Phys. A 51 (2018), no. 50, 505601, 26 pp.

37. Torresblanca-Badillo Anselmo, Zúñiga-Galindo W. A.," Ultrametric diffusion, exponential landscapes, and the first passage time problem," Acta Appl. Math. 157 (2018), 93–116.

38. Bocardo-Gaspar M., Garcia-Compean H., Zúñiga-Galindo W. A., "p-Adic string amplitudes in the limit p approaches to one," J. High Energy Phys. 2018, no. 8, 043, front matter+22 pp.

39. Zúñiga-Galindo W. A., "Non-Archimedean Reaction-Ultradiffusion Equations and Complex Hierarchic Systems," Nonlinearity 31 (2018), no. 6, 2590-2616.

40. Bocardo-Gaspar Miriam, Zúñiga-Galindo W. A., "Local Zeta Functions for Rational Functions and Newton Polyhedra," in Singularities and Foliations. Geometry, Topology and Applications. BMMS 2/NBMS 3, Salvador, Brazil, 2015, Araujo dos Santos, R.N., Menegon Neto, A., Mond, D., Saia, M.J., Snoussi, J. (Eds.), p. 363-386, Springer Proceedings in Mathematics & Statistics.

41. Torresblanca-Badillo A., Zúñiga-Galindo W. A., "Non-Archimedean Pseudodifferential Operators And Feller Semigroups," p-Adic Numbers Ultrametric Anal. Appl. 10 (2018), no. 1, 57-73.

42. Chacón-Cortés L. F., Zúñiga-Galindo W. A., "Heat traces and spectral zeta functions for p-adic Laplacians," Algebra i Analiz, 29 (2017), no. 3, 144–166. Reprinted in St. Petersburg Math. J. 29 (2018), no. 3, 529–544.

43. Zúñiga-Galindo W. A., "Local zeta functions, pseudodifferential operators and Sobolev-type spaces over non-Archimedean local fields," p-Adic Numbers Ultrametric Anal. Appl. 9 (2017), no. 4, 314–335.

44. Veys Willem, Zúñiga-Galindo W. A., "Zeta Functions and Oscillatory Integrals for Meromorphic Functions," Adv. Math. 311(2017), 295–337.

45. Zúñiga-Galindo W. A., "Non-Archimedean white noise, pseudodifferential stochastic equations, and massive Euclidean fields," J. Fourier Anal. Appl. 23 (2017), no. 2, 288–323

46. Chacón-Cortes L. F., Zúñiga-Galindo W. A., "Non-local operators, non-Archimedean parabolic-type equations with variable coefficients and Markov processes," Publ. Res. Inst. Math. Sci. 51 (2015), no. 2, 289–317.

47. Zúñiga-Galindo W. A., "The non-Archimedean stochastic heat equation driven by Gaussian noise," J. Fourier Anal. Appl. 21 (2015), no. 3, 600–627.

48. Zúñiga-Galindo, W. A., "The Cauchy problem for non-Archimedean pseudodifferential equations of Klein–Gordon type," J. Math. Anal. Appl. 420 (2014), no. 2, 1033–1050.

49. Casas-Sánchez O. F., Zúñiga-Galindo, W. A., "p-adic elliptic quadratic forms, parabolic-type pseudodifferential equations with variable coefficients and Markov processes," p-Adic Numbers Ultrametric Anal. Appl. 6 (2014), no. 1, 1–20.

50. León-Cardenal E., Zúñiga-Galindo, W. A., "Local zeta functions for non-degenerate Laurent polynomials over p-adic fields," J. Math. Sci. Univ. Tokyo 20 (2013), no. 4, 569–595.

51. Chacón-Cortes L. F., Zúñiga-Galindo W. A., "Nonlocal Operators, Parabolic-type Equations, and Ultrametric Random Walks," J. Math. Phys. 54, 113503 (2013). 4. Erratum 55, 109901 (2014).

52. Casas-Sánchez O., Zúñiga-Galindo W. A., "Riesz Kernels and Pseudo-differential Operators Attached to Quadratic Forms Over p-adic Fields," p-Adic Numbers, Ultrametric Analysis, and Applications 5 (2013), no. 3, 177-193.

53. Torba S. M., Zúñiga-Galindo W. A., "Parabolic Type Equations and Markov Stochastic Processes on Adeles,"J. Fourier Anal. Appl. 19 (2013), no. 4, 792-835.

54. E. León-Cardenal, Veys, W., Zúñiga-Galindo W.A., "Poles of Archimedean Zeta Functions for Analytic Mappings," J. Lond. Math. Soc. (2) 87 (2013), no. 1, 1-21.

55. Galeano-Peñaloza J., Zúñiga-Galindo W. A., "Pseudo-differential operators with semi-quasielliptic symbols over p-adic fields," J. Math. Anal. Appl. 386 (2012), no. 1, 32–49.

56. Zúñiga-Galindo W.A., " Local zeta functions and fundamental solutions for pseudo-differential operators over p-adic Fields," p-Adic Numbers, Ultrametric Analysis, and Applications 3 (2011), no. 4, 344-358.

57. Segers D., Zúñiga-Galindo W.A., "Exponential sums Along p-adic Analytic Submanifolds," Finite Fields and its Applications 17 (2011), no. 4, 303-316.

58. Moyano-Fernandez J. J., Zúñiga-Galindo W.A., "Motivic Zeta Functions for Curve Singularities," Nagoya Math. J. 198 (2010), 47-75.

59. Rodriguez-Vega J. J., Zúñiga-Galindo W. A., "Elliptic Pseudo-Differential Equations and Sobolev Spaces over p-adic Fields," Pacific J. Math. 246 (2010), no. 2, 407–420.

60. Zúñiga-Galindo W.A., "Local Zeta Functions Supported on Analytic Sets and Newton Polyhedra," Int. Math. Res. Not. IMRN 2009, no. 15, 2855–2898.

61. Rodríguez-Vega, J. J.; Zúñiga-Galindo, W. A., "Taibleson operators, p-adic parabolic equations and ultrametric diffusion," Pacific J. Math. 237 (2008), no. 2, 327-347.

62. Zúñiga-Galindo W.A., "Parabolic Equations and Markov Processes Over p-adic Fields," Potential Anal. 28, (2008), 185-200.

63. Veys W., Zúñiga-Galindo W.A, "Zeta functions associated with polynomial mappings, log-principalization of ideals, and Newton polyhedra," Trans. Amer. Math. Soc. 360 (2008), no. 4, 2205–2227.

64. Zuniga-Galindo W.A., "Multiparametric exponential sums associated with semiquasi-homogeneous polynomial mappings," Finite Fields Appl. 13 (2007), no. 4, 936-945.

65. Zuniga-Galindo W.A., "Decay of Solutions of Wave-Type Pseudo-Differential Equations over p-adic Fields," Publications of RIMS, Vol 42 (2005), 461-479; Errata 44 (2008), no. 1, 45-48.

66. Zuniga-Galindo W.A., "Local zeta function for non-degenerate homogeneous mappings," Pacific J. Math. 218 (2005), no 1, 187-200.

67. Saia M.J., and Zuniga-Galindo W.A., "Local zeta functions for curves, non-degeneracy conditions and Newton polygons," Trans. Amer. Math. Soc. 357 (2005), 59-88.

68. Zuniga-Galindo W.A., "Pseudo-differential equations connected with p-adic forms and Igusa's local zeta functions," Bull. Aust. Math. Soc. 70 (2004), 73-86.

69. Zuniga-Galindo W.A., "On the poles of Igusa's local zeta function for algebraic sets," Bull. London Math. Soc. 36 (2004), 310-320.

70. Zuniga-Galindo W.A., "p-adic oscillatory integrals and Newton polyhedra, In Memoriam of Professor Jairo Charris," Rev. Acad. Colomb. Cienc. Exactas Fís. Nat. Vol 28 (106) (2004), 95-99.

71. Zuniga-Galindo W.A., "Computing Igusa's local zeta functions of univariate polynomials, and linear feedback shift registers," J. Integer Seq. Vol 6(2003), Article 03.3.6.

72. Zuniga-Galindo W.A., "Local zeta functions and Newton Polyhedra," Nagoya Math. J., Vol 172 (2003), 31-58.

73. Zuniga-Galindo W.A. "Fundamental solutions of pseudo-differential operators over p-adic fields," Rend. Sem. Mat. Univ. Padova 109 (2003), 241-245.

74. Zuniga-Galindo, W. A. "Exponential sums along p-adic curves," Finite Fields Appl. 9 (2003), no. 2, 140-149.

75. Zúñiga-Galindo W. A. "Igusa's local zeta functions of semiquasihomogeneous polynomials," Trans. Amer. Math. Soc. 353 (2001), no. 8, 3193-3207.

76. Zúñiga Galindo W. A., "Number of rational points of a singular curve," Proc. Amer. Math. Soc. 126 (1998), no. 9, 2549–2556.

77. Zúñiga-Galindo W. A., "Zeta functions of singular curves over finite fields," Rev. Colombiana Mat. 31 (1997), no. 2, 115-124.

78. Zúñiga Galindo W. A., "Zeta functions and Cartier divisors on singular curves over finite fields," Manuscripta Math. 94 (1997), no. 1, 75-88.

79. Zúñiga Wilson, "On dihedral algebraic function fields," Rev. Colombiana Mat. 27 (1993), no. 3-4, 213-222.

Papers Accepted

80. Zúñiga-Galindo W. A., Chacón-Cortés L.F. "Continuous-Time Quantum Markov Chains And Discretizations Of p-Adic Schrödinger Equations: Comparisons And Simulations." Accepted in the Journal of Statistical Mechanics: Theory and Experiment. arXiv:2508.06712.

Papers Submitted

81. Zúñiga-Galindo W. A., "Wavefunctions localization, and the Wigner's Friend Paradox in a Framework of Discrete-Space Hypothesis." Submitted to Fortschritte der Physik. arXiv:2607.00198.

82. Zúñiga-Galindo W. A., "p-Adic Dirac Equations and the Jackiw-Rebbi Model." Submitted to Journal of Mathematical Physics.

    arXiv:2603.17200.

83. Zúñiga-Galindo W. A., "Critical Organization of Deep Neural Networks, and p-Adic Statistical Field Theories." arXiv:2601.19070. Submitted to the Journal of Physics A: Mathematical and Theoretical.

84. Zúñiga-Galindo W. A. "p-Adic Dirac Equations, Continuous-Time Quantum Walks, and Quantum Networks. "arXiv:2609.04358. Submitted to Chaos, Solitons & Fractals.

Expository and Survey Articles

85. Albis Víctor, Zúñiga-Galindo Wilson, "An elementary introduction to the theory of Igusa local zeta functions," (Spanish) Lect. Mat. 20 (1999), no. 1, 5-33.

86. Leonardo F. Chacón Cortés y Wilson A. Zúñiga Galindo, "Análisis no Arquimediano y Sistemas Complejos," in universo.math.

87. León-Cardenal Edwin, Zúñiga-Galindo W. A., "An introduction to the theory of local zeta functions from scratch," Rev. Integr. Temas Mat. 37 (2019), no. 1, 45–76.

Invited Presentations

1. (6/2026) Local Zeta Functions, String Amplitudes, and Graphs. Sixth International Workshop on Zeta Functions in Algebra and Geometry. 8–12 June 2026, University of Galway, Ireland.

2. (6/2026) Quantum Mechanics, Non-locality, and the Space Discreteness Hypothesis. IARD 2026. The 15th Biennial Conference on Classical and Quantum Relativistic Dynamics of Particles and Fields. 8 - 11 June 2026, Howard University, Washington DC.

3. (2/2026) Continuous-Time Quantum Markov Chains and p-Adic Quantum Mechanics. Probability and Mathematical Physics Seminar at UC Riverside (UCR), February 10, 2026.

4. (12/2025) Statistical Field Theory of Deep Neural Networks: An Ultrametric Approach to Deep Learning. Miami Physics Conference 2025, Dec 12-18, 2025.

5. (11/2025) Continuous-Time Quantum Markov Chains and Discretizations of p-Adic Schrödinger Equations: Comparisons and Simulations. Tercera Escuela sobre la Física de los Materiales Cuánticos, November 4-6, 2025, Tuxtla Gutiérrez, Chiapas, México.

6. (05/2025) p-Adic Quantum Mechanics, Quantum Walks, and the Discreteness of Space.NONLINEARITY, NONLOCALITY AND ULTRAMETRICITY. International Conference on the Occasion of Branko Dragovich 80th Birthday. 26 - 30/05/2025, Belgrade, Serbia.

7. (03/2025) p-Adic numbers, neural networks, deep learning, and statistical field theories. Tercer Encuentro Nacional de Análisis no Arquimediano, Física Matemática y Aplicaciones, Tunja Colombia, March 17-20, 2025.

8. (02/2025) p-Adic Boltzmann Machines. Montréal - Guanajuato Workshop on Probability and Machine Learning, Guanajuato, Mexico, February 25-27, 2025.

9. (11/2024) The p-Adic Schrödinger Equation and the Two-Slit Experiment in Quantum Mechanics. Physics and Its Applications, PHYSICS 2024, Boston, October 21-24, 2024.

10. (06/2024) Quantum Mechanics at the Planck Length. Sixteenth International Conference Application of Mathematics in Technical and Natural Sciences, AMiTaNS24. June 21-26, 2024, Albena, Bulgaria.

11. (04/2024) A Correspondence Between Deep Boltzmann Machines and p-Adic Statistical Field Theories. Spring Eastern Sectional Meeting, Special Session on New Trends in Mathematical Physics, Howard University, Washington, DC, April 6-7, 2024.

12. (04/2024) p-Adic Numbers, Neural Networks, Deep Learning, and Statistical Field Theories.Spring Eastern Sectional Meeting, Special Session on Artificial Intelligence Emergent From Mathematics and Physics, Howard University, Washington, DC, April 6-7, 2024.

13. (03/2024) p-Adic Numbers, Neural Networks, deep learning, and Statistical Field Theory: An introduction. Coloquio de Matemática. Xavierian Pontifical University, Bogotá, Colombia.

14. (04/2023) p-Adic Statistical Field Theory and Deep Boltzmann Machines. Workshop on Applied Analysis and Computation, School of Mathematical and Statistical Sciences, The University of Texas Rio Grande Valley (UTRGV).

15. (11/2022) p-Adic Neural Networks and Quantum Field Theory. International Conference "New Trends in Mathematical Physics, " (November 7–12, 2022). Steklov Mathematical Institute of Russian Academy of Sciences, Moscow, Russia.

16. (11/2022) p-Adic Statistical Field Theory and Deep Belief Networks.5th Annual Meeting, SIAM TX-LA Section, November 4–6, 2022, University of Houston, Houston, TX. USA.

17. (10/2022). Graphs, local zeta functions, log-Coulomb gases, and phase transitions at finite temperature. LAGARTOS =Latin American Geometría Algebraica Real and Tropical Online Seminar.

18. (07/2022) Ultrametric Diffusion, Rugged Energy Landscapes and Transition Networks. Physics and its Applications, July 18-21, San Francisco, CA, United States.

19. (05/2022) Mini course "Introduction to p-adic analysis." CIMPA-CIMAT Research School: p-adic Numbers, Ultrametric Analysis, and Applications, Guanjuato, Mexico.

20. (05/2022) Ultrametric diffusion, rugged energy landscapes and transition networks. Coloquio Virtual FMAT-CIMAT. Mexico.

21. (03/2022) Ultrametricity: perspectives. International Seminar of Mathematics: Juntos UTRGV-UNAL. United States.

22. (11/2021) p-Adic Neural Networks. The 4th Annual Meeting of the SIAM Texas-Louisiana Section (TXLA21). November 5 - 7, 2021.United States.

23. (11/2021) Towards an ultrametric vision of the world. Encuentro Internacional De Matemáticas, EIMAT-2021. Barranquilla, Colombia. Web Conference.

24. (10/2021) Towards an ultrametric vision of the world. **Plenary Speaker**. 54 Congress of the Mexican Mathematical Society. Web Conference, October 17-28, 2021. YouTube video (in Spanish).

25. (05/2021) Non-Archimedean Statistical Field Theory in Eighth International Conference on p-Adic Mathematical Physics and its Applications, Web Conference, May 17-28, 2021. YouTube video.

26. (09/10/2020) p-Adic Numbers, Genomes and Evolution. Web Conference. XV Meeting of the Faculty of Sciences. Pedagogical and Technological University of Colombia, September 11, 2020. Video in Youtube in Spanish.

27. (12/2019) Strings and Quantum Field Theories Over p-adic Spacetimes, Mathematical Physics Seminar, Center for Mathematical Sciences and its Applications, Harvard University, December 2, 2019, Boston, U.S.A.

28. (11/2019) Minicourse: p-Adic Pseudodifferential Equations and Biological Models, in 1er Encuentro en Biología Matemática y Métodos no p-Ádicos, November 11-13, 2019, CIMAT, Merida, México.

29. (11/2019) Ultrametricity in Physics, in CIMAT's Coloquio, November 6, 2019, Guanajuato, México.

30. (10/2019) p-Adic Diffusion and Models of Complex Systems in , Probability Session, 52 Congress of the Mexican Mathematical Society, Monterrey, Mexico, October 21-25, 2019.

31. (10/2019) Reaction-diffusion Equations on Complex Networks and Turing Patterns, via p-Adic Analysis in Seventh International Conference on p-Adic Mathematical Physics and its Applications, Covilhã, Portugal, CMA-UBI, September 30 - October 4, 2019.

32. (06/2019) Mini-Course Introduction to p-Adic Analysis, in p-Adic Analysis, Arithmetic and Singularities, L. Santaló Summer Research School 2019, June 24-29, 2019, Santander, Spain.

33. (06/2019) p-Adic Pseudodifferential Reaction-Diffusion Equations in Mathematical Models, XII Colombian Congress of Mathematics, June10-14, 2019, Popayan, Colombia.

34. (06/2018) Local Zeta Functions and String Amplitudes, Encuentro de Sociedades Matemáticas de Colombia y México, Barranquilla, Colombia, May 30th to June 2nd, 2018.

35. (06/2018) p-Adic Pseudodifferential Equations and Biological Models, Universidad Industrial de Santander, Bucaramanga, Colombia, June 11-13, 2018.

36. (06/2018) p-Adic Pseudodifferential Equations and its Applications, Primera Escuela de verano en aritmética, análisis p-ádicoy física matemática, en honor al Dr. Víctor S. Albis G, Bogotá, Colombia, June 5-8, 2018.

37. (12/2017) p-Adic String Amplitudes and Local Zeta Functions, Algebraic Geometry in Mexico 2017, Oaxaca City and Puerto Escondido, México, November 27th to December 3rd, 2017.

38. (06/2017) String Amplitudes and Local Zeta Functions, Singularity Theory Session. IV Joint Meeting between the Royal Spanish Mathematical Society and the Mexican Mathematical Society, RSME-SMM-2017, Valladolid, Spain.

39. (01/2017) Introduction to the Pseudodifferential Equations Over Non-Archimedean Spaces, Min-course, University of Concepcion, January 16th-25th, 2017, Chile.

40. (10/2016) String Amplitudes and Local Zeta Functions, Congreso Nacional de Geometría Algebraica, Casa Matemática de Oaxaca, Oaxaca City, Mexico, October 3rd to 7th, 2016.

41. (7/2016) Local Zeta Functions and String Amplitudes. ALTENCOA 2016. **Inaugural Conference**, Bucaramanga, Colombia, July 8th to 16th, 2016.

42. (7/2016) Non-Archimedean Equations of Reaction-Diffusion and Complex Systems, Ciclo de Charlas sobre Análisis p-ádico, Geometría Algebraica y Procesos Estocásticos, Bogotá D.C., Colombia, July 26th to 28th, 2016.

43. (7/2016) Non-Archimedean Reaction-Ultradiffusion Equations and Complex Hierarchic Systems, V Congreso Latinoamericano de Matemáticos, Barranquilla, Colombia, July 11th to 15th, 2016.

44. (05/2015) Pseudodifferential Equations on Ultrametric Spaces, Markov Processes, and Models of Complex Systems, Primer Encuentro de Análisis p-ádico y temas afines, May 18th to 20th, 2015, Cali-Colombia.

45. (11/2014) p-Adic and Adelic Parabolic-Type Equations, and Models of Complex Systems. Group Theory, Probability, and the Structure of Spacetime. A Conference in Honor of Professor V. S. Varadarajan's Retirement. UCLA Department of Mathematics, November 7-9, 2014. Los Angeles, U.S.A.

46. (10/2014) Local Zeta Functions, Caloric Equations, and Diffusion in a non-Archimedean Setting. Number Theory Session. XLVII Congress of the Mexican Mathematical Society, Durango, Mexico, October 26-31, 2014.

47. (10/2014) What are the local zeta functions?. Workshop on Algebra, Algebraic Geometry, Algebraic Topology, and Applications. Máximo Gómez Báez University, Ciego de Ávila, Cuba, October 14-18, 2014.

48. (09/2014) Zeta Functions, Heat Equations and Diffusion in an Non-Archimedean Setting. Number Theory Session. III Joint Meeting between the Royal Spanish Mathematical Society and the Mexican Mathematical Society, RSME-SMM-2014, Zacatecas, Mexico.

49. (09/2014) Non degeneracy Conditions, Newton Polyhedra and Zeta Functions. Singularity Theory Session. III Joint Meeting between the Royal Spanish Mathematical Society and the Mexican Mathematical Society, RSME-SMM-2014, Zacatecas, Mexico.

50. (08/2014) Nonlocal Operators, Parabolic-type Equations, and Ultrametric Random Walks, 13th International Conference on p-adic Functional Analysis, University of Paderborn, Paderborn, Germany.

51. (07/2014) Random Walks on Ultrametric Spaces, Parabolic-type Equations, and Models of Complex Systems, UN Encuentro de Matematicas 2014, National University of Colombia, Bogota D.C., Colombia.

52. (04/2013) Non Archimedean Pseudodifferential Equations of Klein-Gordon Type and Quantum Scalar Fields, International Workshop on p-Adic Methods for Modeling Complex Systems, Center for Interdisciplinary Research, University of Bielefeld, Bielefeld, Germany.

53. (08/2013) Oscillatory Integrals for Meromorphic Functions. Brazil-Mexico 1st Meeting On Singularities. Satellite Conference of the Mathematical Congress of the Americas 2013, Queretaro, Mexico.

54. (10/2013) Local Zeta Functions and Parametric Feynman Integrals, Session of Algebraic Geometry, in the XLVI Congress of the Mexican Mathematical Society, Merida, Yucatan, Mexico.

55. (12/2012) Arithmetic and Physics of Complex Systems. Plenary Conference. Algebra, Number Theory, Combinatorics and Applications 2012, ALTENCOA 2012, Bogotá D.C., Colombia.

56. (10/2012) Arithmetic and Physics of Complex Systems. Number Theory Session. 45 Congress of the Mexican Mathematical Society, Queretaro, Mexico, October 28th- November 2nd, 2012.

57. (01/2012) Quadratic Forms, Local Zeta Functions and the Heat Equation Over p-adics. Number Theory Session. II Joint Meeting between

the Royal Spanish Mathematical Society and the Mexican Mathematical Society, RSME-SMM-2012, Torremolinos, Malaga, Spain.

58. (07/2012) P-adic Numbers in Arithmetic and Complex Systems. School of Mathematics for Latin America and the Caribean, EMALCA 2012, Xalapa, Mexico.

59. (03/2012) Introduction to the Theory of Local Zeta Functions (Minicurse). Algebraic Structures, their representation and applications in geometry and non-associative models. Research School CIMPA-UNESCO-MESR-MICINN-COLOMBIA, Cartagena, Colombia.

60. (07/2011) Polynomial Congruences and Poincaré Series Over p-adic Manifolds. **Plenary Conference** at XVIII Colombian Congress of Mathematics, Bucaramanga, Colombia, 11-15 July 2011.

61. (06/2010) Zeta functions, Pseudo-differential Operators Over Non-Archimedean Local Fields and Complex Systems, ALTENCOA4-2010, Algebra, Number Theory, CCombinatoricsand Applications, Tunja, Colombia.

62. (08/2009) A p-adic Multivariate Version of Igusa's Stationary Phase Method, XVII Colombian Congress of Mathematics, Cali, Colombia.

63. (07/2009) Motivic Zeta Functions for Curve Singularities. Algebraic Geometry Session. I Joint Meeting between the Mexican Mathematical Society and the Royal Spanish Mathematical Society, Oaxaca City, Mexico.

64. (07/2009) A p-adic Multivariate Version of Igusa's Stationary Phase Method. Number Theory Session. I Joint Meeting between the Mexican Mathematical Society and the Royal Spanish Mathematical Society, Oaxaca City, Mexico.

65. (10/2008) Local Zeta supported on Analytic Sets, Session of Geometry and Algebraic Geometry, XLI Congress of the Mexican Mathematical Society, Valle de Bravo, Mexico.

66. (10/2008) Digital Signatures Based on Elliptic Curves, AMA2008, Querétaro, México.

67. (07/2008) Local Zeta FunFunctions Supported Analytic Subsets and Newton Polyhedra, in 10th International Workshop on Real and Complex Singularities. São Carlos, Brazil.

68. (07/2008) Mini-Course on Local Zeta Functions for Analytic Mappings, Institute of Mathematical Sciences and Computation, University of São Paulo, Brazil.

69. (06/2008) Diffusion and Parabolic Equations over p-adics, Department of Mathematics, National University of Colombia, Bogotá.

70. (06/2008) Some Recent Results on Local Zeta Functions Attached to Analytic Mappings, Department of Mathematics, National University of Colombia, Bogotá.

71. (05/2008) Zeta Functions Attached to Analytic Mappings, Seminar of Algebraic Geometry, Institute of Mathematics, National Autonomous University of Mexico, UNAM, Campus Morelia.

72. (04/2008) On the Construction of Candidates to One-Way Functions Based on the Computation of the Number of Solutions of Polynomial Congruences, International Congress of Engineering, Autonomous University of Queretaro, México.

73. (04/2008) Parabolic Equations over p-adics, Seminar of Applied Analysis,Querétaro, Mexico.

74. (02/2008) What is Public-key Cryptography? Mathematical Week at CINVESTAV, Mexico City, Mexico.

75. (06/2007) Zeta Functions of Analytic Mappings, Principalization of Ideals, and Newton Polyhedra, Algebraic Geometry Seminar, Complutense University, Madrid, Spain.

76. (09/2006) Local Zeta Functions for Polynomial Mappings and Multivariate Exponential Sums, Collaborative Number Theory Seminar, Graduate Center, The City University of New York, CUNY, New York, U.S.A.

77. (03/2006) (Mini-course) Local zeta functions and p-adic pseudo-differential equations, Department of Mathematics, National University of Colombia, Bogotá.

78. (10/2005) Parabolic Pseudo-differential Equations and Markov Processes Over p-adic Fields, Third Southern Florida Analysis Seminar, Florida International University, Miami, U.S.A.

79. (04/2005) Decay of Wave-type Pseudo-differential Equations Over p-adic Fields, Florida Atlantic University, Analysis Seminar, Boca Raton, U.S.A.

80. (06/2004) Computing Igusa's Local Zeta Functions, and an Application to Cryptography, Number Theory and Algebraic Geometry Seminar, K. U. Leuven- U. Gent, Leuven, Belgium.

81. (06/2004) Igusa's Local Zeta Functions and p-adic Pseudo-differential Equations, Number Theory and Algebraic Geometry Seminar, K. U. Leuven – U. Gent, Leuven, Belgium.

82. (01/2004) Pseudo-differential Equations Connected With p-adic Forms and Igusa's Local Zeta Functions, AMS Special Session on Arithmetical Algebraic Geometry, Joint Meetings AMS-MAA in Phoenix, USA.

83. (05/2002) Computation of Local Zeta Functions, Trees, and One-way Functions, 2002 Illinois Number theory Conference, University of Illinois at Urbana-Champaign, USA.

84. (04/2000) Digital Signatures Based on Elliptic Curves, Seminar of the Department of Mathematics and Computer Science, Barry University.

85. (01/2000) (Mini-course) An Elementary Introduction to Igusa's Local Zeta Functions, (Summer Program), Institute of Mathematics and Computation, São Paulo University, ICMC-USP, campus São Carlos, Brazil.

86. (01/2000) Complex Powers and Newton Polyhedra, II MiniWorkshop on Singularities, Geometry and Differential Equations, Institute of Mathematics and Computation, São Paulo University, ICMC-USP, campus São Carlos, Brazil.

87. (07/1999) Local Zeta Functions and Newton Polyhedra, Encontro de Algebra Commutativa e Geometria Algebrica, ALGA99, Campinas University, Campinas, Brazil.

88. (02/1998) Igusa's Local Zeta Functions (p-adic Stationary Phase Formula), Graduate Center, City University of New York, CUNY, USA.

89. (07/1997) Igusa's Local Zeta Functions of Semiquasi-homogeneous Polynomials, Encontro de Algebra Commutativa e Geometria Algebrica, ALGA97, Institute of Pure and Applied Mathematics, IMPA, Rio de Janeiro, Brazil.

90. (07/1996) Zeta Functions of Singular Curves Over Finite Fields, XIV Brazilian School of Algebra, Institute of Pure and Applied Mathematics, IMPA, Rio de Janeiro, Brazil.

91. (06/1996) On the Zeta Functions Associated to the Group of Cartier Divisors, Seminar of Algebra, Institute of Pure and Applied Mathematics, IMPA, Rio de Janeiro, Brazil.

92. (06/1995) On the Number of Rational Points of a Singular Curve, Seminar of Algebra, Institute of Pure and Applied Mathematics, IMPA, Rio de Janeiro, Brazil.

Postdocs Supervised

Dr. Samuel Estala Arias. Postdoctoral Fellow supported by Conacyt, Mexico (2014-2015).

Ph.D. Dissertations Supervised

1. Chayapuntika Indoung, "Pattern Formation in Quantum and Classical Hierarchical Cellular Neural Networks," University of Texas Rio Grande Valley, 2026.

2. Baboucarr Dibba, "p-Adic Cellular Neural Networks With Delay," University of Texas Rio Grande Valley, 2025.

3. Nathaniel Mayes, "p-Adic Quantum Mechanics, Infinite Potential Wells, and Continuous-Time Quantum Walks," University of Texas Rio Grande Valley, 2025.

4. Carlos Alberto Garcia Bibiano, "p-Adic FitzHugh-Nagumo Systems and Turing Patterns," Cinvestav-I.P.N, Mexico, 2023.

5. Brian Andres Zambrano Luna, "p-Adic Cellular Neural Networks and Applications." Cinvestav-I.P.N, Mexico, 2023.

6. Angela Rocío Fuquen Tibatá, "p-Adic Open String Amplitudes and Quantum Field Theory." Cinvestav-I.P.N, Mexico, 2023.

7. Víctor Antonio Aguilar Arteaga, " Contributions to the Study of the Pseudodifferential Equations over Ultrametric Spaces." Cinvestav-I.P.N, Mexico, 2019.

8. María Luisa Martínez Mendoza, "Quantum field theory of scalar fields on a p-adic space-time." Cinvestav-I.P.N, Mexico, 2019. Other advisor Dr. José Antonio Vallejo.

9. Edilberto Arroyo Ortiz, "Construction of p-adic Quantum Fields in the Framework of White Noise Analysis." Cinvestav-I.P.N, Mexico, 2019.

10. Miriam Bocardo Gaspar, Local Zeta Functions and p-adic String Amplitudes." Cinvestav-I.P.N, Mexico, 2017.

11. Adriana Alexandra Albarracín Matilla, "Igusa's Local Zeta Functions and Exponential Sums for Arithmetically Non Degenerate Polynomials." Cinvestav-I.P.N, Mexico, 2017. Other advisor Dr. E. León Cardenal.

12. Anselmo Torresblanca Badillo, "Ultrametric Diffusion, Energy Landscapes, Feller Semigroups, and Pseudodifferential Operators Over p-adics." Cinvestav-I.P.N, Mexico, 2017.

13. Leonardo Fabio Chacón Cortés, "Non-local operators, Pseudodifferential Equations of Parabolic Type With Variable Coefficients and Markov Processes over p-adics." Cinvestav-I.P.N, Mexico, 2014.

14. Oscar Francisco Casas Sánchez, "Local Zeta Functions, Functional Equations and Pseudodifferential Operators Over p-adic Fields." National University of Colombia, Bogota, 2014.

15. Edwin León Cardenal, "Local Zeta Functions for Analytic Mappings, Non-degeneracy Conditions, and Newton Polyhedra." Cinvestav-I.P.N, Mexico, 2013.

16. Jeanneth Galeano Peñaloza, "Pseudo-differential Equations With Quasi-elliptic Symbols Over p-adic Fields ." Cinvestav-I.P.N, Mexico, 2012.

17. John J. Rodriguez Vega, "Local Zeta Functions and Pseudo-Differential Equations Over p-adic Fields," National University of Colombia, Bogota, 2010. Other advisor Dr. V. Albis González.

Master's Theses in Mathematics Guided

1. Emmanuel Abelardo Roque Jiménez, "A-padic Model of Biological Evolution" Cinvestav-I.P.N, Mexico, 2021.

2. Brian Andrés Zambrano Luna, "Local Zeta Functions for Graphs." Cinvestav-I.P.N, Mexico, 2020. Other advisor Dr. Edwin León Cardenal.

3. Victor Antonio Aguilar Arteaga, "Local Zeta Functions via the Stationary Phase Formula." Meritorious Autonomous University of Puebla, Mexico, 2014.

4. Edwin León Cardenal, "An Algorithm for Computing the Local Zeta Function Attached to an Hyperelliptic Curve." National University of Colombia, Bogotá, 2008.

Master's Theses in Computer Science Guided

1. Jaime Arcila and Juan Lambrano, "Evaluating Security of Networks using S.A.T.A.N and I.S.S." Autonomous University of Bucaramanga, Colombia, 1999.

2. Miguel Cadena and Juan Martinez, "A Windows Package for Digital Signatures Using Elliptic Curves." Autonomous University of Bucaramanga, Colombia, 2000.

3. Claudia Solano and Fernando Perez, "A Windows Package for Generating Cryptographically Strong Primes." Autonomous University of Bucaramanga, Colombia, 2000.

4. Francis Castellanos Ariza, "Developing and Testing of Security Policies on Web Servers using REFEREE." Autonomous University of Bucaramanga, Colombia, 2001.

5. Desiderio Espinosa and Hernando Gonzales, "Computing the Number of Rational Points of Elliptic Curves over Finite Fields." Autonomous University of Bucaramanga, Colombia, 2001.

6. Flavio Pinto Siabatto, "Fractal Segmentation in Studying Sandstone Porosity ."Autonomous University of Bucaramanga, Colombia, 2001.

7. Giovanny Mendoza, Rafael Vázquez, " Design and Implementation of a Windows Application for Private Mail Using Lucas Sequences." Autonomous University of Bucaramanga, Colombia, 2002.

8. Cesar Guerrero, " Security Evaluation of Intranets." Autonomous University of Bucaramanga, Colombia, 2002.

Organized Conferences

1. (Jointly with Tristan Hubsch) Special Session on New Trends in Mathematical Physics in 2024 Spring Eastern Sectional Meeting, Howard University, Washington, DC, April 6-7, 2024.

2. CIMPA-CIMAT Research School: p-Adic Numbers, Ultrametric Analysis, and Applications. Guanajuato, Mexico, Mayo 23-31, 2022 (jointly with Trond Digernes).

3. Fifth International Workshop on Zeta Functions In Algebra And Geometry, May 2-6, 2022, Nice, France (jointly with Alejandro Melle Hernandez, Adam Parusiński, Wim Veys, Ann Lemahieu).

4. (Conference's Chairman) Eighth International Conference on p-Adic Mathematical Physics and its Applications, Web Conference, May 17-28, 2021. YouTube Channel of the Conference.

5. First Encounter in Mathematical Biology and p-Adic Methods, (Jointly with Yuriria Cortés Poza, IIMAS, UNAM-Yucatán, and J. Rogelio Pérez Buendía, CIMAT-Mérida.) November 11-13, 2019, CIMAT, Merida, Mexico.

6. p-Adic Analysis, Arithmetic and Singularities, L. Santaló Summer Research School 2019 (Jointly with A. Melle Hernandez, C. Galindo Pastor, J. J. Moyano Fernández), June 24-29, 2019, Santander, Spain.

7. Mini-Workshop on p-Adic Mathematical Physics (Jointly with Sergii Torba), Querétaro, May 13th-15th, 2019.

8. (Conference's Chairman) Sixth International Conference on p-adic Mathematical Physics and Its Applications, Mexico.p-adics2017, Mexico City, October 23rd-27th, 2017.

9. (Jointly with Benjamin Klopsch, Willem Veys, Christopher Voll)Fourth International Workshop on Zeta Functions in Algebra and Geometry, Bielefeld, Germany, May 29th to June 2nd 2017.

10. Organizer of the Special Session of Non-Archimedean Analysis (jointly with Przemyslaw Górka, Alexander Quintero, Enrique Reyes García), in V Congreso Latinoamericano de Matemáticos, Barranquilla, Colombia, July 11th to 15th, 2016.

11. Organizer of the Number Theory Session. XLVII Congress of the Mexican Mathematical Society, Durango, Mexico, October 26th-31th, 2014.

12. (Jointly with A. Campillo, A. Melle-Hernandez, W. Veys) Third International Workshop on Zeta Functions in Algebra and Geometry, Guanajuato, Mexico, September 8th to 12th, 2014.

13. Organizer of the First International Conference on Models of Complex Hierarchic Systems and Non-Archimedean Analysis, April 29th to May 3rd 2013, Mexico City, Mexico.

14. Organizer of the Number Theory Session in the XLV Congress of the Mexican Mathematical Society, October 28th-November 2nd 2102, Queretaro, Mexico.

15. (Jointly with Javier Cilleruelo) Number Theory Session in the Joint Meeting between the Real Sociedad Matematica Española and Sociedad Matematica Mexicana, 17th-20th January 2012. Torremolinos, Spain.

16. (Jointly with Michael Porter, Sergii Torba) New Trends in Analysis, Geometry and Physics. Celebrating the 50th Anniversary of the Department of Mathematics of CINVESTAV-I.P.N., 18th-19th August 2011, Queretaro, Mexico.

17. (Jointly, A. Campillo, A. Melle-Hernandez, W. Veys) Second International Workshop on Zeta Functions in Algebra and Geometry, Palma de Mallorca, Spain, May 3rd to 7th, 2010.

18. Number Theory Session, in XLII National Congress of the Mexican Mathematical Society, October 12th to 16th, 2009, Zacatecas, Mexico.

19. Number Theory Session, in First Joint Meeting between the Mexican Mathematical Society and the Royal Spanish Mathematical Society, July 22nd to 24th, 2009, Oaxaca City, Mexico.

20. (Jointly with M. Porter and V. Kravchenko), Seminar of Applied Analysis, April 14th, 2008, Querétaro, México.

21. (Jointly with M. Porter and V. Kravchenko), AMA2008 - Actualidad de las Matemáticas Aplicadas, Encuentro Nacional- October 6th to 8th, 2008, Querétaro, México.

22. (Jointly with A. Melle-Hernandez, W. Veys),International Workshop on Zeta Functions in Algebra and Geometry, June 25th to 29th, 2007, Segovia, Spain.

Teaching

I have more than twenty-seven years of university teaching experience in Colombia, Mexico, and the United States; see Appointments above. Course lists for UTRGV (2021–2026) and for earlier positions follow.

The University of Texas Rio Grande Valley (2021–2026)

At UTRGV, teaching evaluations are reported as an "H-score" (0–5 scale). Across 15 evaluated sections from Spring 2021 through Spring 2026, my average H-score is **4.10**, with several graduate and upper-division courses rated above 4.4.

Graduate courses taught:

- MATH 8329, Analytic Number Theory (Spring 2026).
- MATH 6331, Algebra I (Spring 2025, two sections).
- MATH 8331, Abstract Algebra (Spring 2025).
- MATH 8698, Interdisciplinary Research (Spring 2025; Fall 2024).
- MATH 8335, Deep Learning (Spring 2024).
- MATH 8398, Interdisciplinary Course (Spring 2023; Fall 2022).
- MATH 6352, Analysis I (Spring 2021).
- MATH 9301/9601/9901/9902, Dissertation and Doctoral Research supervision (continuous, Spring 2021–Summer 2026).

Undergraduate courses taught:

- MATH 2346, Mathematics for Electrical and Computer Engineering (Fall 2025; Fall 2024).
- MATH 2318, Linear Algebra (Fall 2024; Fall 2023; Spring 2022).
- MATH 4355, Topology (Fall 2023).
- MATH 4390, Mathematics Project (Fall 2025).
- MATH 3341, Differential Equations (Spring 2023; Spring 2022).
- MATH 2415, Calculus III (Fall 2022).
- MATH 2414, Calculus II (Spring 2023).
- MATH 4342, Complex Variables (Fall 2021).

Earlier positions

CINVESTAV, Mexico (2007–2020). Graduate courses only (the Department of Mathematics admits only master's and doctoral students): Functional Analysis; Real Analysis; Fundamentals of Computer Science; Topics in Number Theory; Algebraic Geometry; p-Adic Analysis; Topics in Local Zeta Functions.

Barry University, U.S.A. (2001–2007). A six-course annual load spanning service through upper-division courses: Introduction to Computers; Pre-calculus; Introduction to Topology; Computer Security; Data Communications; Applied Cryptography; Computer Forensics; Introduction to Mathematical Modeling.

Colombia (1996–2000). Undergraduate and graduate courses in mathematics and computer science at the Industrial University of Santander and the Autonomous University of Bucaramanga.

Professional Service

The items below cover my years at UTRGV (2021–2026). Service earlier in my career appears under Editorships, Grants, Organized Conferences, and Books Edited above, which span my full career.

Editorial boards. Member of the editorial boards of six international journals; see the Editorships section above.

Conference organization. Chair or co-organizer of twenty-two international conferences, schools, and workshops; see the Organized Conferences section above.

External doctoral committees.

- (06/2026) Ph.D. committee for Ángel Morán Ledezma, Karlsruhe Institute of Technology, Germany.
- (01/2024) Ph.D. committee for Vincenzo Parisi, University of Camerino, Italy.

Refereeing. Referee for numerous international journals, including: Proceedings of the Royal Society A; Pacific Journal of Mathematics; Forum Mathematicum; Annals of Physics; Journal of Mathematical Physics; Nuclear Physics B; Journal of Mathematical Analysis and Applications; Journal

of Pseudo-Differential Operators and Applications; p-Adic Numbers, Ultrametric Analysis, and Applications; Proceedings of the Steklov Institute of Mathematics; Physica A: Statistical Mechanics and its Applications; Chaos, Solitons and Fractals; Machine Learning: Science and Technology; Neurocomputing; Scientific Reports; Reports on Mathematical Physics; Physica Scripta; Expositiones Mathematicae; and Boletín de la Sociedad Matemática Mexicana.

Research seminar. Founder and organizer of the bi-weekly Virtual Research Seminar on Non-Archimedean Analysis and Mathematical Physics, which involves graduate students and external collaborators.

University Service

The University of Texas Rio Grande Valley, 2021–2026.

Endowed chair and professorship selection committees.

- (2022-2023) Fred W. and Frances H. Rusteberg Faculty Fellowship in Science and Technology Committee.
- (2021-2022) Myles and Sylvia Aaronson Professorship Committee.
- (2021-2022) Houston Endowment Chair in Science, Mathematics & Technology Committee.
- (2021-2022) Carlos and Stephanie Manrique de Lara Professorship in Science Committee.
- (2021-2022) President's Endowed Professorship in the College of Science and College of Engineering and Computer Science Committee.

Promotion, tenure, and faculty review.

- (2025-2026) Tenure Committee.
- (2024-2025) Post-Tenure Review Committee.

Faculty recruitment.

- (2025-2026), (2022-2023), (2021-2022) Faculty Search Committee.

Graduate program service.

- (2025-2026) Ph.D. defense committees for Julio Cesar Paez and Roberto Arias.
- (2025-2026) Ph.D. candidacy exam committee for Erika Olivares.
- (2025-2026), (2024-2025) Ph.D. committee, School of Mathematical and Statistical Sciences.
- (2023-2024) Ph.D. candidacy exam committee for Jeffery Opuku.
- (2025-2026), (2024-2025), (2023-2024), (2022-2023) Chair of the Analysis Comprehensive Examination.

Curriculum Development

- (2024-UTRGV) MATH 8335, Deep Learning (new doctoral course).
- (2005-Barry University) Computer Forensics.
- (2005-Barry University) Introduction to Mathematical Modeling.
- (2005-Barry University) Applied Cryptography.
- (2001-Barry University) Computer Security.